The Political Economy of Debt Maturity and Sovereign Default Risk*

Preliminary and Incomplete

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October 21, 2026

Abstract

This paper studies how political conflict shapes sovereign borrowing, default and debt maturity. In our model, generations of workers and retirees elect governments every period. Workers favor fiscal discipline, whereas retirees favor public spending financed by labor taxes and debt. Default risk stems from a stochastic cost of default. Our main theoretical contribution is to show that, even without income risk, the maturity structure is determinate: short- and long-term debt are issued, and maturity shortens during debt crises. This contrasts with normative theory prescribing that governments facing no income risk should remain passive in long-term bond markets. The mechanism combines debt dilution with the fiscal discipline of the workers. Anticipating that the next government will dilute legacy debt, workers favor long-term issuance because it crowds out future public spending less than short-term issuance. Using high-frequency polling data for stressed euro area countries, we find that higher support for anti-austerity parties is associated with higher sovereign yield spreads, particularly on bonds maturing after the next election, lower debt issuance and shorter maturities of new issuance.

Keywords: sovereign debt, sovereign default, debt maturity, government bond yield spread, political process, election, election poll

JEL Classification: D72, E43, F34, F41, H63

*We would like to thank seminar and conference participants at the University of Helsinki; Swiss Macro Workshop; Swiss Society for Economics and Statistics Annual Meeting; and the UniNE-KOF-UNIBAS PhD Macro Workshop for their valuable comments. Financial support from the Swiss National Science Fundation (Grant 215537) is gratefully acknowledged.

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1 Introduction

Public debt in the euro area remains at historically high levels. At the end of 2025, general government debt stood at 87.8 percent of GDP in the euro area and exceeded 100 percent of GDP in Greece, Italy, France, Belgium, and Spain. At the same time, the political landscape has shifted. Across Europe, the vote share of far-right parties, most of which are populist, has risen in the most recent national elections. This paper studies theoretically and empirically how political conflict shapes sovereign borrowing, default and debt maturity.

On the theoretical side, we develop an electoral-competition model of sovereign debt in which overlapping generations of voters elect governments that choose taxes, public spending, default, and the issuance of debt. We show that political conflict pins down the maturity structure of public debt. In particular, both short- and long-term debt is issued in equilibrium, and the maturity of issued debt shortens during debt crises. On the empirical side, we test the model's predictions using high-frequency polling data for the stressed euro area countries during the European sovereign debt crisis. We find that rising support for anti-austerity parties is associated with higher sovereign bond yield spreads, especially on bonds maturing after the next election, with lower debt issuance, and with a shorter maturity of newly issued debt.

Our model builds on Song et al. (2012) and Müller et al. (2016) to include default risk and a choice between short- and long-term debt. The model abstracts from ideology and income risk, while default risk stems from a stochastic cost of default as in Aguiar et al. (2019). The accumulation of sovereign debt arises from an intergenerational conflict between young voters (i.e. workers), who favor fiscal discipline, and old voters (i.e. retirees), who favor high public expenditures financed through labor income taxes and debt accumulation. Subsequent generations of young and old voters elect a new government in every period. Incumbent governments have neither commitment to future policies nor to current debt repayment. We focus on Markov-perfect equilibria and characterize them through generalized Euler equations (GEEs), which capture how current choices affect the policies of future governments.

Our main result is that the maturity structure is determinate and that long-term debt is issued in equilibrium. This contrasts with the benchmark result of Aguiar et al. (2019) that, absent income risk, a sovereign should remain passive in long-term bond markets and actively manage only short-term debt. The key to our result is a dilution motive that appears in the GEEs. Because long-term bonds issued in the past are still outstanding when the current government borrows, an expansion of total borrowing lowers the value of this legacy debt. Holding constant the value of the outstanding debt portfolio, legacy long-term debt therefore

raises the marginal revenue from new issuance. As a result, each government borrows more, spends more, and taxes less than in an otherwise identical economy holding only short-term debt of equal market value.

The fiscal discipline of the young shapes debt maturity. Young voters favor issuing long-term debt because, compared with issuing only short-term debt, it crowds out less of next period's public spending, which they will enjoy when old. They anticipate that the next government will find it optimal to dilute the inherited long-term debt, which raises public spending in the next period. This future dilution benefit compensates for the lower issuance revenue of long-term bonds, whose price already reflects the anticipated dilution. Issuing only short-term debt would eliminate dilution, but such a policy is not time consistent. The model also predicts that the maturity of debt shortens during debt crises.

We test these predictions using high-frequency polling data for Greece, Ireland, Italy, Portugal, and Spain during the European sovereign debt crisis. We use electoral support for anti-austerity parties as a proxy for a perceived decline in the future cost of default. The reason is that these parties opposed austerity programs and, once in office, could weaken the political commitment to repay. Consistent with the model, higher support for anti-austerity parties is associated with higher sovereign bond yield spreads, particularly for bonds maturing after the next election. We further show that higher polling support is associated with lower overall debt issuance and with a shorter maturity of newly issued debt. These patterns mirror those documented for emerging economies during financial crises as in Broner et al. (2013), but they arise in advanced economies and in response to political risk.

Taken together, our results show that political conflict is an independent driver of sovereign debt dynamics. Even without income risk or ideological shocks, it generates debt accumulation and default risk, and the prospect of political change shapes both the cost and the maturity of sovereign debt. Our findings also show that debt maturity is not neutral for the distribution of resources across generations. This suggests that debt management choices, and institutional arrangements that limit dilution, carry political and distributional consequences that go beyond their usual role in managing refinancing risk. These considerations are especially relevant for the euro area today given historically high levels and populist parties gaining ground.

The remainder of the paper is organized as follows. Section 1.1 reviews the related literature. Section 2 presents the model. Section 3 characterizes the equilibrium and the maturity structure. Sections 4 and 5 present the quantitative and empirical analyses, respectively. The Appendix contains the microfoundation of the political process, proofs and additional details about the quantitative and the empirical analyses.

1.1 Literature Review

This paper contributes to the literature on time-consistent fiscal policy chosen through repeated voting (Krusell and Rios-Rull, 1999; Hassler et al., 2003, 2005; Song, 2011, 2012). We extend the electoral-competition model of Song et al. (2012) and Müller et al. (2016) by introducing default risk and a choice between short and long-term debt. In the canonical sovereign default model of Eaton and Gersovitz (1981), Aguiar and Gopinath (2006) and Arellano (2008), the government borrows to smooth consumption in the face of a stochastic endowment. In our framework, the incentive to accumulate debt arises instead from political conflict. In this respect, we complement the literature that identifies political conflict as a driver of public debt accumulation (Cukierman and Meltzer, 1989; Alesina and Tabellini, 1990; Battaglini and Coate, 2008; Yared, 2010; Azzimonti, 2011), and we study how this motive interacts with default risk and the maturity structure of debt.

Our analysis also relates to the literature on the political economy of sovereign default. Closest to our work are Scholl (2017) and Prein and Scholl (2021), who introduce endogenous political turnover through probabilistic voting into the canonical sovereign default model. Our framework differs in three respects. First, we abstract from ideology since voting is driven only by the economic consequences of fiscal policy. Second, we introduce long-term debt alongside short-term debt. Third, we assume deterministic income in our small open economy, and default risk stems from a stochastic cost of default, as in Aguiar et al. (2019). Other studies relate default risk to the political leaning of governments (Cotoc et al., 2025), tax revolts (Arce et al., 2024), legislative constraints (Azzimonti and Mitra, 2023), and inequality and redistribution (Dovis et al., 2016; Andreasen et al., 2019; Hermann and Scholl, 2023). Relative to this literature, we show that political conflict alone, without income risk or ideological shocks, generates debt accumulation and default risk, and we characterize how it shapes the maturity structure of sovereign debt.

We consequently contribute to the literature on the optimal maturity structure of sovereign debt. Under commitment, Angeletos (2002) shows that the maturity structure can be used to complete markets, but the required positions are extreme and amount to several multiples of GDP (Buera and Nicolini, 2004; Faraglia et al., 2010).[1] Accounting for default risk, Arellano and Ramanarayanan (2012) and Niepelt (2014) find that the maturity shortens as default risk increases.[2] In these models, income is stochastic, and the government trades off hedging against low income realizations, which calls for longer maturities, against the dilu-

[1] In the model of Lucas and Stokey (1983) under incomplete markets, Debortoli et al. (2017) show that the trade-off between hedging and the incentive to manipulate interest rates yields a nearly flat maturity structure. Accounting for default risk, Kiiashko (2022) shows that the optimal maturity is decaying.

[2] Perez (2017) obtains a similar result when lenders do not observe the government's willingness to repay.

tion incentives created by long-term debt. Without income risk, Aguiar et al. (2019) show that the sovereign should remain passive in long-term bond markets and actively manage only short-term debt. Our model also abstracts from income risk, yet it predicts that government issues long-term debt in equilibrium and that the maturity of debt shortens in bad times. Diluting legacy debt allows the next government to accumulate debt faster, financing higher spending and lower taxes, which current young voters value.[3] Chatterjee et al. (2026) additionally find that long-term debt can be optimal under credible protection of creditors and limits on discretionary spending.

Our study relates to the empirical literature of sovereign debt in times of stress. For eleven emerging economies over 1990-2009, Broner et al. (2013) show that the risk premium on long-term bonds exceeds that on short-term bonds, and that the gap widens during crises while issuance shifts toward shorter maturities. Consistent with this, the maturity of new issuance falls when spreads rise (Arellano and Ramanarayanan, 2012; Perez, 2017) and during recessions (Bai et al., 2017). Our findings for the stressed euro area countries mirror those of Broner et al. (2013), but the trigger is political rather than financial. Long-term spreads react more than short-term spreads to the rise of anti-austerity parties, and issuance shifts toward short-term debt. Political factors have previously been linked to sovereign risk. In emerging economies, spreads are higher where governments are less stable (Scholl, 2017) and in countries that elect the left more often (Cotoc et al., 2025). Closest to our data, Prein and Scholl (2021) use Greek election polls to show how support for anti-bailout parties and the 10-year spread evolved during the bailout programs. Political instability around elections and government popularity booms also predict crises (Bussière and Mulder, 2000; Herrera et al., 2019). None of these studies, however, examines the maturity dimension.

Our characterization relies on a generalized Euler equation (GEE). Without commitment, policy is the outcome of a game between successive decision makers (Krusell et al., 2002; Krusell and Smith, 2003; Krusell et al., 2010). Klein et al. (2008) characterize differentiable Markov-perfect equilibria through a GEE that captures how today's choices affect future policy rules, and use it to compute equilibria. Song et al. (2012) and Müller et al. (2016) apply this approach to electoral-competition models. We further extend this characterization with default risk and multiple maturities, and therefore complement the work of Mateos-Planas et al. (2026).

[3]Long-term debt exposes legacy creditors to dilution. Because future governments ignore how new issuance lowers the value of outstanding bonds, long-term creditors are exposed to dilution (Hatchondo and Martinez, 2009; Chatterjee and Eyigungor, 2012), which accounts for most of the default risk in calibrated models (Hatchondo et al., 2016). Dilution can also be self-fulfilling and generate multiple equilibria (Aguiar and Amador, 2020; Stangebye, 2020). It can be mitigated by seniority clauses (Chatterjee and Eyigungor, 2015), by commitment to a borrowing path (Hatchondo et al., 2020), or by committing one period ahead to the default decision and the repurchase price (Mateos-Planas et al., 2025).

2 Model

Following Song et al. (2012), we study a small open economy populated by overlapping generations of two-period-lived voters, classified as young and old. The economy has an infinite horizon, time is discrete, and a new government is elected in every period. In the original framework of Song et al. (2012), sovereign debt is restricted to a one-period maturity and governments are fully committed to honoring inherited obligations. We relax both assumptions, which we view as particularly relevant in the context of the European sovereign debt crisis. Specifically, we extend the framework to allow for long-term debt and limited commitment to repay outstanding debt, as is standard in the sovereign default literature. The remainder of this section describes the model environment in detail.

2.1 Preferences

Agents consume a private good, c, and a domestic public good, g, provided by the incumbent government. Young agents supply labor, h, which enters production of the private good, whereas old agents finance private consumption through their savings. The expected lifetime utility of a young agent and the utility of an old agent are given by

$$EU_y = \log(c_y - v(h)) + \theta \log(g) + \beta E[U_o'|X] - d\phi \tag{1}$$

$$U_o = \log(c_o) + \lambda\theta \log(g) - d\phi, \tag{2}$$

where $\beta \in (0,1)$ is the discount factor, U_o' denotes the continuation utility of a young agent, and θ and $\lambda\theta$ are the preference weights for public relative to private goods for young and old agents, respectively. Labor disutility $v(h)$ enters young agents' utility in the Greenwood et al. (1988) form and we assume that $v(h) = h^{1+1/\xi}/(1+1/\xi)$ where $\xi > 0$ is the Frisch elasticity of labor supply. This preference structure closely follows Müller et al. (2016), with the addition of the government's default decision, represented by the indicator $d \in \{0,1\}$. In the event of default, $d = 1$, young and old voters suffer a stochastic utility cost $\phi \in \mathcal{N} = [0, \overline{\phi}] \subset \mathbb{R}^+$ where $0 < \bar{\phi} < \infty$ as in Müller et al. (2019). We assume that ϕ is $i.i.d.$ according to the distribution function $F(\phi)$ that is differentiable on the interior of $\mathcal{N}$, with density $f(\phi)$. Further, $E[\cdot|X]$ denotes expectations conditional on $X \equiv \{F, \langle D, G\rangle\}$, where D and G denote the equilibrium policy functions for default and public expenditure, respectively, as formally defined in Definition 1.

Taking government policy as given, young agents choose labor supply and savings to

maximize their lifetime utility, subject to the lifetime budget constraint,

$$c_y + \frac{c'_o}{R} = (1-\tau)wh \quad \text{with} \quad c'_o = Rs,$$

where $\tau \in (-\infty, 1]$ denotes a proportional tax on labor income, s is the savings of young agents and $R > 1$ the gross world interest rate.[4] The optimal choice of labor supply is then $H(\tau) = [(1-\tau)w]^{\xi}$. Consequently, all remaining optimal choices are proportional to after-tax labor income $A(\tau) = [(1-\tau)w]^{1+\xi}$. Optimal savings and consumption allocations are given by

$$s(\tau) = \frac{\beta}{(1+\beta)(1+\xi)}A(\tau), \; c_y(\tau) = \frac{1+\xi(1+\beta)}{(1+\beta)(1+\xi)}A(\tau), \text{ and } c'_o(\tau) = \frac{\beta R}{(1+\beta)(1+\xi)}A(\tau).$$

2.2 Production

The private good is produced using the Cobb-Douglas production technology $Y = QK^{\alpha}H^{1-\alpha}$ with capital K and labor H as inputs and $\alpha \in (0,1)$. Q corresponds to the total factor productivity. Capital is perfectly mobile and depreciates fully at the end of every period. We assume a competitive factor market such that the optimal level of capital and the pre-tax wage are respectively

$$K = K(R,H) = \left(\alpha \frac{Q}{R}\right)^{\frac{1}{1-\alpha}} H \quad \text{and} \quad w = (1-\alpha)Q^{\frac{1}{1-\alpha}}\left(\frac{\alpha}{R}\right)^{\frac{\alpha}{1-\alpha}}.$$

To simplify the exposition, we normalize Q so that $w = 1$. This implies that $H(\tau) = [(1-\tau)]^{\xi}$ and $A(\tau) = [(1-\tau)]^{1+\xi}$.

2.3 Public Good Provision

The public good can be provided only by the incumbent government, which has access to international financial markets. The government finances the public good through the proportional labor income tax, τ, short-term debt, $b'_s \in \mathbf{B} \equiv [0, \bar{b}]$, and long-term debt $b'_l \in \mathbf{B}$.[5] The debt vector is denoted by $b' \equiv (b'_s, b'_l)$.

Short-term debt consists of one-period bonds with unit price $q_s(b')$ that pays a coupon

[4] Similar to Song et al. (2012) and Müller et al. (2016), we assume that the government cannot tax consumption and wealth.

[5] We assume that the upper bound on debt, $\bar{b}$, is sufficiently large, for instance the natural borrowing limit $1/(R-1)$, such that it never binds in equilibrium. Further, since debt is reset to $b = 0$ after a default, zero is a natural lower bound for the state space. In our calibrated economy, this lower bound does not bind either.

$\kappa \geq 0$ at maturity. Long-term debt consists of two-period bonds with unit price $q_l(b')$ that pay the same coupon κ in each period. Let b_l denote the stock of outstanding two-period bonds issued in the previous period and maturing next period. Let b_s denote the total debt due in the current period, comprising one-period bonds issued in the previous period and two-period bonds issued two periods earlier. Given the inherited debt vector, $b = (b_s, b_l)$, the government's budget constraint is

$$q_l(b')b'_l + q_s(b')(b'_s - b_l) = g + (1+\kappa)b_s + \kappa b_l - \tau H(\tau). \tag{3}$$

The government bonds are held by international creditors with deep pockets. These creditors are risk-neutral, competitive, and have access to the risk-free gross world interest rate R. Equilibrium bond prices $q_l(b')$ and $q_s(b')$ therefore satisfy the condition that lenders break even in expectation.

The government cannot commit to repay the inherited debt. If it chooses to default, all the outstanding debt is written off, $b = 0$. However, the current young and old voters suffer the default cost $\phi > 0$.

2.4 Political process

A new government is elected in every period by the current young and old voters. Thus, each government can only commit to the current fiscal policy. An election consists of a competition between two office-seeking candidates who announce fiscal policy platforms. We assume politicians have no ideological stance, so the chosen platform maximizes the probability of winning the election and reflects voters' preferences exclusively.

Under a standard probabilistic voting setup, the policy platforms of the two candidates converge in equilibrium.[6] Conditional on current debt repayment, $d = 0$, both candidates aim to maximize the political objective function

$$O(\tau, g, b') = (1-\omega)EV_y(\tau, g, b') + \omega V_o(g), \tag{4}$$

where $\omega \in (0, 1)$ denotes the political weight of old relative to young voters and the indirect utility functions of the young and old voters conditional on repayment are, respectively,

$$EV_y(\tau, g, b') = \Psi_y + (1+\beta)\log(A(\tau)) + \theta \log(g) \tag{5}$$

[6]The microfoundation of the political objective function can be found in Appendix A. It largely follows Song et al. (2012), but extends their framework by incorporating the default decision, d, as an additional element of the government's policy platform.

$$+ \beta\lambda\theta \int_{\mathcal{N}} \left[(1 - D(b', \phi')) \log G(b') + D(b', \phi') \left(\log(G(0)) - \frac{\phi'}{\lambda\theta} \right) \right] \mathrm{d}F(\phi')$$

$$V_o(g) = \Psi_o + \lambda\theta \log(g). \tag{6}$$

They are obtained by substituting the optimal consumption, $c_y(\tau)$ and $c'_o(\tau)$, optimal labor supply, $H(\tau)$, and $d = 0$ into (1) and (2), writing out the conditional expectation $E[\cdot|X]$. Ψ_y and Ψ_o are terms that are independent of d, τ, g, and b' and therefore do not affect the equilibrium fiscal policy.

When the incumbent government chooses to default, $d = 1$, debt cancels from the government budget constraint (3), which then simplifies to $q_l(b')b'_l + q_s(b')b'_s = g - \tau H(\tau)$. Further, the political objective function additionally accounts for the cost of default, $O(\tau, g, b') - \phi$, which enters the utility of the young and old voters additively. Note that in our model politicians are purely rent-seeking with no ideology of their own. This means that the utility cost reflects the societal economic cost of default rather than the idiosyncratic political ideology of each candidate.

3 Political Equilibrium

Fiscal policy in equilibrium arises from a dynamic game among successive governments. As in the previous literature, we focus on Markov-perfect equilibria, under which their strategies depend only on payoff-relevant state variables. Since accumulated savings do not affect the policy preferences of old agents, the relevant state vector consists solely of the inherited debt position, b, and the default cost realization, ϕ. Let $\Omega \equiv \mathbf{B}^2 \times \mathcal{N}$ denote the state space:

Definition 1. *A Markov-perfect Political Equilibrium (MPE) is a short-term bond price schedule* $q_s : \mathbf{B}^2 \to [0, (1+\kappa)/R]$*, a long-term bond price schedule* $q_l : \mathbf{B}^2 \to [0, (\kappa + (1+\kappa)/R)/R]$ *and a 5-tuple of functions* $\langle D, B, G, T \rangle$ *where* $D : \Omega \to \{0, 1\}$ *is a default rule,* $d = D(b, \phi)$*,* $B : \mathbf{B}^2 \to \mathbf{B}^2$ *is the vector of bond rules for short-term debt* $b'_s = B_s(b)$ *and long-term debt* $b'_l = B_l(b)$*,* $G : \mathbf{B}^2 \to \mathbb{R}^+$ *is the public expenditure rule* $g = G(b)$ *and* $T : \mathbf{B}^2 \to (-\infty, 1]$ *is the tax rule* $\tau = T(b)$ *such that the fiscal policy rule is optimal and time-consistent,*

$$D(b, \phi) = \arg \max_{d \in \{0,1\}} (1 - d)W(b) + d[W(0) - \phi] \tag{7}$$

$$\langle B(b), G(b), T(b) \rangle = \underset{(b'_s, b'_l) \in \mathbf{B}^2, g \geq 0, \tau \leq 1}{\arg\max} (1 - \omega) E V_y(\tau, g, b') + \omega V_o(g) \tag{8}$$

$$s.t. \quad (3), (5), (6) \text{ and } b \in \mathcal{B}(\phi),$$

where

$$W(b) \equiv (1-\omega)EV_y(T(b), G(b), B(b)) + \omega V_o(G(b))$$
$$\mathcal{B}(\phi) \equiv \{b \in \mathbf{B}^2 : W(b) \geq W(0) - \phi\},$$

and the bond price functions satisfy

$$q_s(b') = \int_{\mathcal{N}} (1 - D(b', \phi')) \frac{1+\kappa}{R} \mathrm{d}F(\phi'), \tag{9}$$
$$q_l(b') = \frac{1}{R} \int_{\mathcal{N}} (1 - D(b', \phi')) \left[\kappa + q_s(B(b'))\right] \mathrm{d}F(\phi'). \tag{10}$$

The government decides in (7) whether to honor the inherited debt by comparing the value of repayment, $W(b)$, and default, $W(0) - \phi$. In the repayment state, it then chooses the optimal short- and long-term bond issuance, public expenditure, and tax rate for the current period according to (8). We require that the fiscal policy rule is time-consistent: the same policy functions $\langle D, B, G, T\rangle$ appear both on the left– and right-hand side of the maximization problems. In the default state, the fiscal policy rule is simply $\langle B(0), G(0), T(0)\rangle$, capturing that the outstanding debt cancels from the government budget constraint (3).

The assumption of *i.i.d.* default cost leads to the price of the bonds being dependent on b' only. Furthermore, the long-term bond price can be expressed as a function of the short-term bond price since $q_l(b') = \frac{1}{R}\int_{\mathcal{N}}(1-D(b',\phi'))[\kappa+\frac{1}{R}\int_{\mathcal{N}}(1-D(B(b'),\phi''))(1+\kappa)dF(\phi'')]dF(\phi') = \frac{1}{R}\int_{\mathcal{N}}(1-D(b',\phi'))[\kappa+q_s(B(b'))]dF(\phi')$.

Henceforth, we focus on equilibria in which the optimal choice b' is on the interior of $\mathcal{B}(\bar{\phi})$ and the policy rules $\langle B, G, T\rangle$ and prices are differentiable at b'. We label them differentiable Markov-perfect political equilibria (DMPE), which are characterized as follows:

Proposition 1. *A DMPE is characterized by the following system of functional equations:*
1. The default rule

$$D(b, \phi) = \begin{cases} 1 & \text{if } \phi < \Phi(b) \\ 0 & \text{else} \end{cases}, \tag{11}$$

where the default threshold satisfies $\Phi(b) \equiv W(0) - W(b)$*;*
2. The bond prices

$$q_s(b) = (1 - F(\Phi(b))) \frac{1+\kappa}{R}, \tag{12}$$

$$q_l(b) = (1 - F(\Phi(b))) \frac{1}{R} [\kappa + q_s(B(b))]; \tag{13}$$

3. The budget constraint

$$q_l(B(b))B_l(b) + q_s(B(b))(B_s(b) - b_l) = G(b) + (1+\kappa)b_s + \kappa b_l - T(b)H(T(b)); \tag{14}$$

4. The intra-temporal optimality condition

$$\frac{(1-\omega)(1+\beta)(1+\xi)}{A(T(b))} = \frac{\theta[1+\omega(\lambda-1)]}{G(b)}[1 - e(T(b))], \tag{15}$$

where $e(\tau) \equiv -(\partial H(\tau)/\partial\tau)/(\tau/H(\tau))$ *denotes the tax elasticity of the labor supply;*
5. The inter-temporal optimality conditions (generalized Euler equations)

$$\frac{\theta[1+\omega(\lambda+1)]}{G(b)}\left[q_s(b') + \frac{\partial q_s(b')}{\partial b'_s}(b'_s - b_l) + \frac{\partial q_l(b')}{\partial b'_s}b'_l\right] \tag{16}$$
$$= -(1-\omega)\beta\lambda\theta\left[\frac{1 - F(\Phi(b'))}{G(b')}\frac{\partial G(b')}{\partial b'_s} - f(\Phi(b'))\frac{\partial \Phi(b')}{\partial b'_s}\left(\log(G(b')) - \log(G(0)) + \frac{\Phi(b')}{\lambda\theta}\right)\right],$$

$$\frac{\theta[1+\omega(\lambda+1)]}{G(b)}\left[q_l(b') + \frac{\partial q_l(b')}{\partial b'_l}b'_l + \frac{\partial q_s(b')}{\partial b'_l}(b'_s - b_l)\right] \tag{17}$$
$$= -(1-\omega)\beta\lambda\theta\left[\frac{1 - F(\Phi(b'))}{G(b')}\frac{\partial G(b')}{\partial b'_l} - f(\Phi(b'))\frac{\partial \Phi(b')}{\partial b'_l}\left(\log(G(b')) - \log(G(0)) + \frac{\Phi(b')}{\lambda\theta}\right)\right],$$

where $b' = B(b) \in \text{int}(\mathcal{B}(\bar{\phi}))$, $b'_s = B_s(b)$, *and* $b'_l = B_l(b)$.

The assumption that $b' \in \text{int}(\mathcal{B}(\bar{\phi}))$ serves two purposes. First, it eliminates the non-differentiability of $F(\phi)$ at the border of its support $\mathcal{N}$. Second, it allows us to focus on the empirically relevant case of an interior default probability, $F(\Phi(b)) \in (0,1)$, which is the case most pertinent to our quantitative analysis in the context of the European sovereign debt crisis.

Equation (11) states that the binary default rule $D(b,\phi) \in \{0,1\}$ can be reformulated as a threshold rule $\Phi(b)$ as in the canonical sovereign debt model. It assumes that at indifference, the government chooses to repay. The two bond prices in Equations (12) and (13) can then be rewritten accordingly, focusing only on states with repayment. Equation (14) is simply the equilibrium budget constraint. Equation (15) captures the tradeoff between the marginal cost of taxation and the marginal benefit of public good provision, taking into account the tax elasticity of the labor supply $e(\tau)$. This optimality condition reflects the intratemporal

political conflict between young and old voters: the latter are retired from the labor force and would like to set labor income taxes at the top of the tax Laffer curve, $\overline{\tau} = \arg\max_{\tau}\{\tau H(\tau)\}$, in order to maximize current public good provision.

Equations (16) and (17) are the generalized Euler equations (GEEs) for short- and long-term debt issuance, respectively. The government trades off the marginal utility from current public good provision for all voters against the marginal expected utility from future public good provision for currently young voters. The GEEs reflect the inter-generational political conflict emphasized in Song et al. (2012): young voters impose fiscal discipline as they internalize that debt accumulation crowds out future public good provision and increases the default probability under the next government. This is captured by the derivatives of the public expenditure rule, $G(b')$, and the default threshold, $\Phi(b')$, with respect to debt issuance on the right-hand side of the GEEs.[7] In opposition to young voters' fiscal discipline, old voters prefer to maximize the public good provision today and would consequently support setting taxes at $\overline{\tau}$ and debt at the top of the borrowing Laffer curve $\overline{b}(b_l) = \arg\max_{b' \in \mathbf{B}^2}\{q_s(b')(b'_s - b_l) + q_l(b')b'_l\}$. Yet, like young voters, they internalize that debt accumulation increases default risk, a consideration reflected in the price derivatives present on the left-hand side of the two GEEs.

3.1 Equilibrium debt maturity structure

How does the inter-generational political conflict differ between short- and long-term debt issuance? For each government, diluting the value of the outstanding long-term debt held by legacy lenders is both optimal and time-consistent. Holding constant the value of the outstanding debt, a portfolio with legacy debt, $b_l > 0$, increases the marginal revenue of newly issued debt, $q_l(b')b'_l + q_s(b')(b'_s - b_l)$, through the reduction in the value of legacy debt, $q_s(b')b_l$. As a result, the current government issues more debt and entertains higher public expenditure and lower taxes. Although these policies increase the welfare of current voters, the associated cost is shifted to future generations. Consequently, debt maturity affects the nature of inter-generational political conflict. Young voters prefer a portfolio with long-term debt issuance because it crowds out next period's government expenditures less compared to issuing only short-term debt. The young internalize that the next government will find it optimal to dilute the inherited long-term debt, thereby increasing future public expenditures.

[7] In our framework with defaultable debt, the standard disciplining effect captured by $\partial G(b')/\partial b'_s$ and $\partial G(b')/\partial b'_l$ only operates in future states with repayment and therefore softens as debt increases and the repayment probability $1 - F(\Phi(b'))$ falls. In addition, young voters internalize that debt accumulation increases the future default probability, captured by $\partial\Phi(b')/\partial b'_s$ and $\partial\Phi(b')/\partial b'_l$. This effect adds to fiscal discipline if, at the default cost threshold $\Phi(b')$, repayment is more attractive than default for the young.

This mitigates the lower issuance revenues associated with long-term bonds relative to short-term bonds.

The following proposition establishes formally that this mechanism determines the maturity structure of debt issuance. In particular, the government finds it optimal to issue both short-term and long-term debt.

Proposition 2. *In a DMPE,* $B_l(b) > 0$.

The proof of Proposition 2 exploits that a policy where only short-term bonds are traded violates the GEEs. The proposition shows that each government optimally leaves the next government a debt portfolio with a positive amount of long-term debt. Issuing only short-term debt as in Aguiar et al. (2019), avoids future debt dilution but is not time-consistent in our political equilibrium. Further, the result applies independent of whether the economy inherits legacy long-term debt, b_l.

Figure 1 illustrates the proposition using our calibrated economy with parameter values as described in Table 1. In each panel, the corresponding equilibrium function varies with b_s along the horizontal axis and is plotted for three different values of b_l. The top-right panels show that, conditional on repayment, short- and long-term debt accumulation converges to an interior fixed point $b^* = B(b^*)$, where $b_l^* = 0.32$ is associated with the red solid line in the figure. Further, following a default where debt is reset to $b = 0$, the government issues a portfolio with strictly positive short- and long-term debt (blue line). Figure 2 displays the same DMPE, but plots the policy rules against the market value $\pi(b)$ on the horizontal axis. The second panel shows that the expenditure rule is falling in $\pi(b)$ for a given b_l. However, holding constant $\pi(b)$, a debt portfolio with more legacy debt b_l mitigates the crowding out effect of debt on public expenditures. That illustrates the rationale for young voters' support for long-term debt issuance.

3.2 Comparison with alternative environments

In the presence of income risk, Arellano and Ramanarayanan (2012) show that the hedging benefits of long-term debt can offset the adverse pricing effects associated with debt dilution, making long-term debt a relevant instrument for debt management. By contrast, our characterization of the debt maturity structure in Proposition 2 does not depend on income risk. Even in its absence, the government actively relies on long-term debt issuance. This result differs from Aguiar et al. (2019).

To illustrate the limits of Proposition 2, the next definition contrasts our political equilibrium with the optimal time-consistent fiscal policy of a dynastic government with an infinite

horizon. Following Farhi and Werning (2007), the dynastic government internalizes the welfare of future generations by assigning weight ψ^{t+1}, with $\psi \in [0,1)$, to the expected utility of each generation born in period $t = -1, 0, \ldots, \infty$.

Definition 2. *A Markov-perfect Political Dynasty Equilibrium (MPE*) is a fiscal policy rule that is optimal and time-consistent,*

$$D(b,\phi) = \arg\max_{d\in\{0,1\}} dW^*(b) + (1-d)[W^*(0) - \phi] \tag{18}$$

$$\begin{aligned} W^*(b) = &\max_{b'\in\mathbf{B}^2, g\geq 0, \tau\leq 1} \beta\theta\lambda\log(g) + \psi\left[(1+\beta)\log(A(\tau)) + \theta\log(g)\right] \\ &+ \psi\int_{\mathcal{N}} [(1-D(b',\phi'))W^*(b') + D(b',\phi')[W^*(0)-\phi']]\, dF(\phi') \\ s.t. \quad &(3), (9), (10), \text{ and } b \in \mathcal{B}^*(\phi), \end{aligned} \tag{19}$$

where $\mathcal{B}^*(\phi) \equiv \{b \in \mathbf{B} : W^*(b) \geq W^*(0) - \phi\}$.

The crucial difference to the MPE in Definition 1 is that (19) can be expressed as a Bellman equation in terms of the value function W^*. Let $\psi = (1-\omega)\beta/\omega$, then a differentiable MPE* (DMPE*) is characterized by (11)–(15), and the modified GEEs

$$\frac{\theta[\beta\lambda+\psi]}{G(b)}\left[q_s(b') + \frac{\partial q_s(b')}{\partial b'_s}(b'_s - b_l) + \frac{\partial q_l(b')}{\partial b'_s}b'_l\right] = -\psi[1-F(\Phi(b'))]\frac{\partial W^*(b')}{\partial b'_s}, \tag{20}$$

$$\frac{\theta[\beta\lambda+\psi]}{G(b)}\left[q_l(b') + \frac{\partial q_l(b')}{\partial b'_l}b'_l + \frac{\partial q_s(b')}{\partial b'_l}(b'_s - b_l)\right] = -\psi[1-F(\Phi(b'))]\frac{\partial W^*(b')}{\partial b'_l}, \tag{21}$$

where $b' = B(b) \in \text{int}(\mathcal{B}^*(\bar{\phi}))$, $b'_s = B_s(b)$, and $b'_l = B_l(b)$. Since $\psi = (1-\omega)\beta/\omega$, the only difference between a DMPE* and a DMPE in Proposition 1 is the right-hand side of the GEEs. The derivative of the continuation value in repayment states, $W^*(b')$, captures the marginal effect on the welfare of current and all future generations rather than only the currently young. In equilibrium, we can equivalently express $W^*(b') = \bar{W}^*(\pi(b'), b'_l)$. By the envelope theorem, debt issuance affects $W^*(b')$ only through its impact on the debt value, $\pi(b')$, in the government budget constraint. Hence, $\partial\bar{W}^*(\pi(b'), b'_l)/\partial b'_l = 0$, so that issuing a portfolio with long-term debt is no longer beneficial at the margin and $b'_l = 0$ remains a DMPE* candidate.

While it is beyond the scope of this section to formally prove that there indeed exists a DMPE* where only short-term debt is traded, Figure D.1 displays a numerical solution. We use the same calibration of parameters as in Figure 1 and set the weight $\psi = (1-\omega)\beta/\omega$.[8]

[8]Only the value of the discount factor is recalibrated ($\beta = (0.952)^{30}$) such that the DMPE* has the same

The figure shows that, consistent with Aguiar et al. (2019), there exists a DMPE* where the government manages long-term debt passively until maturity. In particular, it neither issues nor repurchases long-term debt and instead relies exclusively on short-term debt as an active debt-management instrument. The intuition is that, conditional on repayment, consumption is deterministic, so sovereign default risk is the only source of intertemporal trade-offs. Short-term debt is preferred because its price fully internalizes the marginal effect of borrowing on default risk, thereby aligning incentives period by period. Long-term debt introduces no additional margin of control. Instead, trading long-term debt tightens the government's budget constraint through adverse price adjustments without generating any offsetting benefit in terms of repayment incentives. In our political equilibrium, by contrast, long-term debt issuance affects not only default incentives but also the future path of fiscal policy, rendering long-term debt trades non-redundant. The crucial difference is that future generations do not enter the government's objective function. As a result, debt maturity acquires an additional political-economy role, which explains the departure from Aguiar et al. (2019) established in Proposition 2.

4 Quantitative Analysis

In this section, we calibrate the model and evaluate its ability to replicate a set of empirical moments related to debt maturity and sovereign bond spreads. We then use the calibrated model to analyze how shocks to the perceived cost of default affect bond spreads and debt issuance across the maturity spectrum.

4.1 Calibration

The model is calibrated to stressed euro area countries (Greece, Ireland, Italy, Portugal, and Spain) over the period 2000–2010, which spans the creation of the euro and the run-up to the sovereign debt crisis.

To approximate the equilibrium functions B, G, T, and Φ that solve the system of functional equations in Proposition 1, we employ a projection method based on Chebyshev collocation. We focus on the DMPE that arises as the limit of a finite-horizon economy. Accordingly, the solution algorithm begins by solving the finite-horizon political equilibrium via backward induction. It is initialized in the terminal period, where young voters live for only one remaining period, and proceeds backward until the collocation coefficients converge. We then verify that the limiting solution of the finite-horizon economy coincides with the

average debt-to-GDP ratio in the stationary equilibrium.

Table 1: Calibration

Parameter	Value	Target	Data	Model
A. Data-Based				
α	0.399	Capital income share		
R	$(1.047)^{30}$	Capital output ratio (annualized)		
Q	3.394	Domestic wage (normalization)		
ω	0.500	Voter turnout for old (55+) in EU Parliament		
λ	0.713	Effective labor tax rate		
ξ	0.613	Effective labor tax rate at top of Laffer curve		
B. Model-Based				
β	$(0.996)^{30}$	Debt output ratio (annualized)	75.76	76.65
θ	0.373	Short-term debt share	38.27	38.43
$\overline{\phi}$	0.800	Term Premium long vs. short-term debt	0.04	0.01
η	2.500	Short-term interest rate spread	0.49	0.54

solution to the infinite-horizon problem. Appendix C provides details of the numerical algorithm.

Table 1 reports the values of all model parameters. The parameters in Panel A are calibrated directly from empirical counterparts, while those in Panel B are chosen to match the stationary-equilibrium moments reported in the table. As in Song et al. (2012), a model period corresponds to 30 years. The Frisch elasticity of labor supply is set to $\xi = 0.613$, implying that the peak of the Laffer curve occurs at $\overline{\tau} = 0.62$, following Trabandt and Uhlig (2011). We set $\omega = 0.5$ to match the voter turnover among older voters in the 2009 European Parliament elections and choose $\lambda = 0.713$ to replicate an effective labor income tax rate of 0.25.

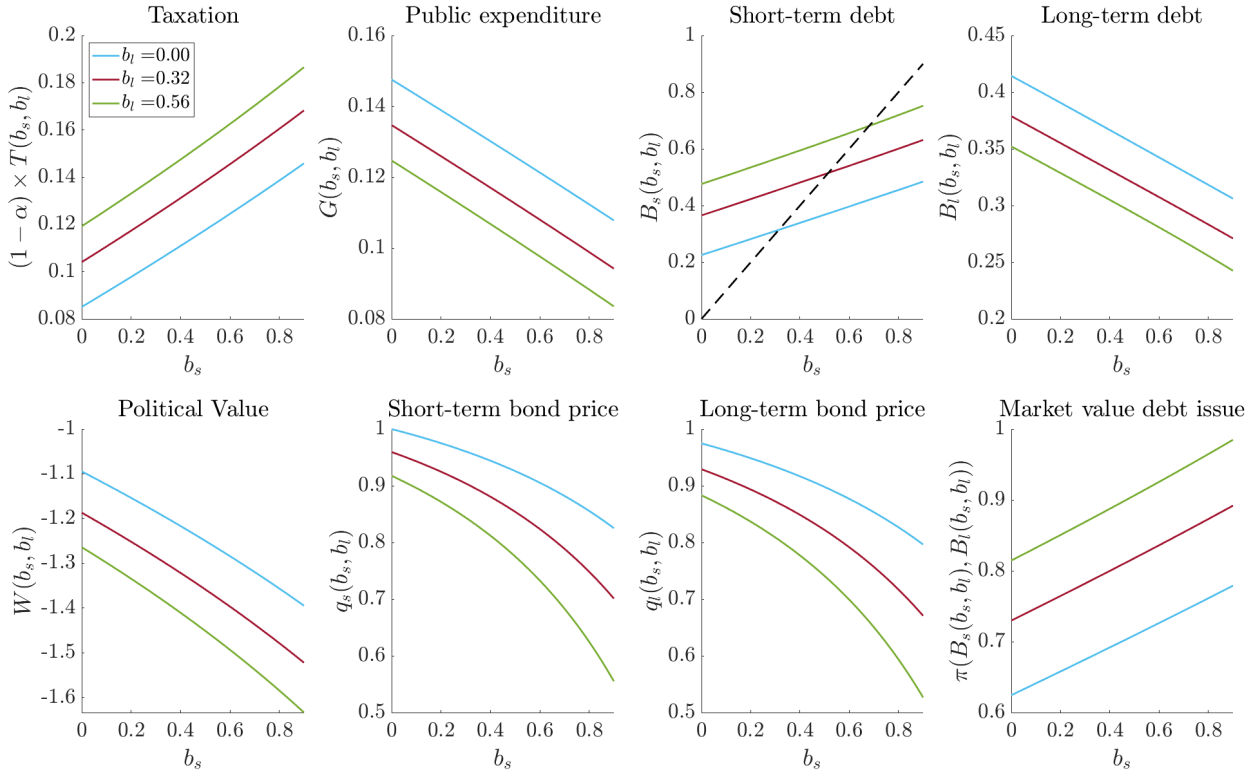
Using data from AMECO, $1 - \alpha$ corresponds to the mean adjusted wage share in the sample. We set $R = (1.047)^{30}$, which comes from multiplying the capital income share by the average annual capital-output ratio of 3. Given this, we further normalize the total factor productivity $Q = (1-\alpha)^{\alpha-1}(\alpha/R)^{-\alpha}$. We finally set $1 + \kappa = R$ meaning that the coupon payment corresponds to the net interest rate. Consequently, $q_s(b) = (1 - F(\Phi(b)))$ and $q_l(b) = (1 - F(\Phi(b)))\frac{1}{R}[R - 1 + q_s(B(b))]$.

Regarding specific moments matching, we set β to match the average debt-to-GDP ratio and θ to match the short-term debt share. We assume $\phi \in [0, \overline{\phi}]$ follows a truncated exponential with rate η. We set $\overline{\phi}$ to match the term premium between the average 2-year and 1-year government bond yields and η to match the average 1-year interest rate spread relative to Germany.

4.2 Main Policy Functions and Prices

Figure 1 illustrates the equilibrium functions of the calibrated political equilibrium. The main policy functions are the tax policy, $T(b_s, b_l)$, the public expenditure policy, $G(b_s, b_l)$, and the short-term and long-term debt policies, $B_s(b_s, b_l)$ and $B_l(b_s, b_l)$. We also report the political value, $W(b_s, b_l)$, the short-term and long-term bond prices, $q_s(b_s, b_l)$ and $q_l(b_s, b_l)$, and the market value of debt issuance, $\pi\big(B(b_s, b_l)\big)$, where the market value of a debt portfolio with $1 + \kappa = R$ is defined as

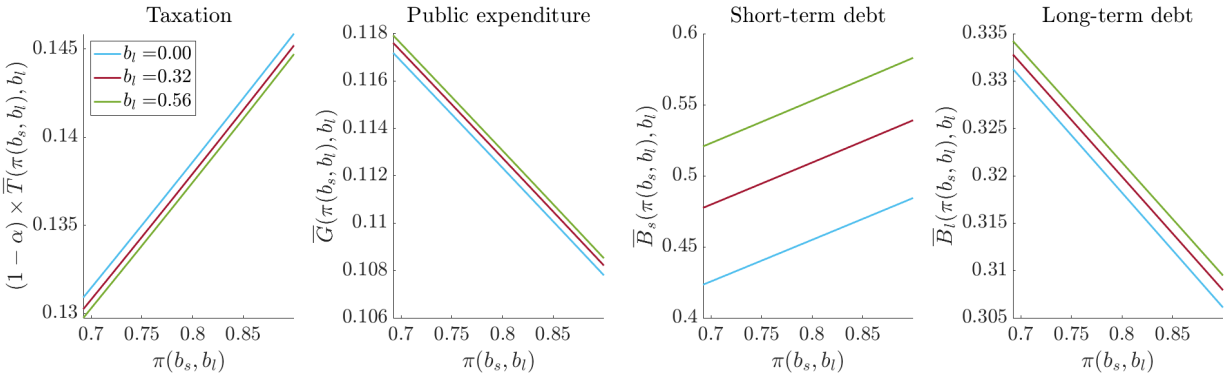
$$\pi(b_s, b_l) \equiv b_s + b_l \frac{1}{R}\big[R - 1 + q_s\big(B(b_s, b_l)\big)\big].$$



Note: The figure depicts the rule for taxation, $(1-\alpha)\times T(b_s, b_l)$, public expenditure, $G(b_s, b_l)$, short-term debt, $B_s(b_s, b_l)$, long-term debt, $B_l(b_s, b_l)$, the objective function, $W(b_s, b_l)$, the short-term bond price, $q_s(b_s, b_l)$, the long-term bond price, $q_l(b_s, b_l)$ and the market value of debt issuance, $\pi(B(b_s, b_l))$ as a function of b_s. The different lines correspond to different values of b_l.

Figure 1: Policy Functions, Values and Prices as a Function of b_s

The figure plots these objects against short-term debt, b_s, for three levels of long-term debt, b_l. Taxation, short-term debt issuance, and the market value of debt issuance increase with both b_s and b_l, whereas public expenditure, long-term debt issuance, political value, and both bond prices decrease with both. The monotonicity of the value and the bond



Note: The figure depicts the rule for taxation, $(1-\alpha)\times\overline{T}(\pi(b_s,b_l),b_l)$, public expenditure, $\overline{G}(\pi(b_s,b_l),b_l)$, short-term debt, $\overline{B}_s(\pi(b_s,b_l),b_l)$, and long-term debt, $\overline{B}_l(\pi(b_s,b_l),b_l)$ as a function of $\pi(b_s,b_l)\equiv b_s+b_l\frac{1}{R}[R-1+q_s(B(b_s,b_l))]$. The different lines correspond to different values of b_l.

Figure 2: Policy Functions as a Function of $\pi(b_s,b_l)$

prices follows the standard argument that more debt implies higher debt servicing costs and therefore lower value of repayment and higher default risk. Two further features are worth noting. First, the government substitutes across maturities. A larger stock of either type of debt leads it to issue more short-term and less long-term debt. Second, both types of debt mean-revert. The slope of B_s with respect to b_s is below one, so each line crosses the 45-degree line once from above, at a level of short-term debt that increases with b_l.

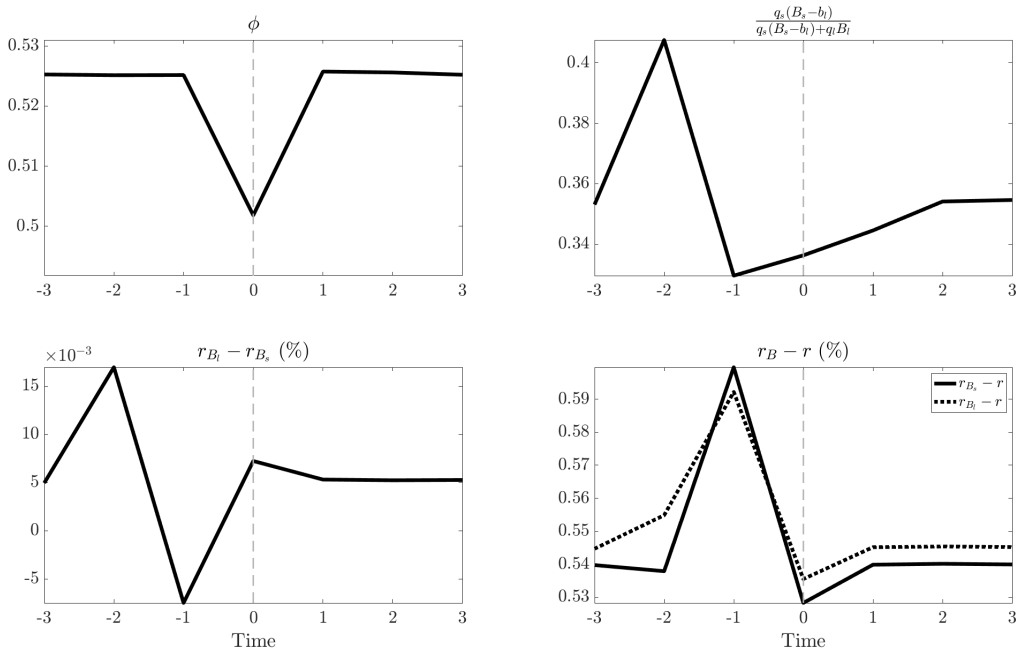
Figure 2 plots the same objects against the market value of the inherited portfolio, $\pi(b_s,b_l)$. For each b_l, we recover the level of b_s consistent with a given market value and denote the resulting functions with a bar, e.g. $\overline{T}(\pi,b_l)\equiv T(b_s,b_l)$ with b_s such that $\pi(b_s,b_l)=\pi$. This change of variable is important. Moving across lines at a given point on the horizontal axis shifts the composition of the portfolio towards long-term debt while holding its market value constant. The monotonicity with respect to π is the same as with respect to b_s in Figure 1, although the issuance policies are considerably flatter. The monotonicity across b_l, however, is reversed for taxation, public expenditure and long-term debt issuance. For a given market value, a portfolio tilted towards long-term debt leads to higher public expenditure, lower taxation and more issuance of both short-term and long-term debt. This reflects the debt dilution motive. The larger the outstanding stock of long-term debt, the larger the share of the cost of new borrowing borne by legacy long-term bondholders.

The composition of the debt portfolio is not neutral. For a given market value, short-term and long-term debt have different effects on public expenditure and taxation. This non-neutrality is what pins down the maturity structure in the model, as shown in Proposition 2. Young voters favor issuing long-term debt because it raises future public expenditure and lowers taxes. They anticipate the next government's debt dilution motive, which com-

pensates for the lower bond revenue per unit of long-term debt relative to short-term debt.

4.3 Default-Cost Shock

Our model generates a rich set of predictions for the behavior of bond spreads and debt issuance across maturities. In this section, we first analyze their response to a decrease in the expected cost of default and then confront the model's predictions with the data.

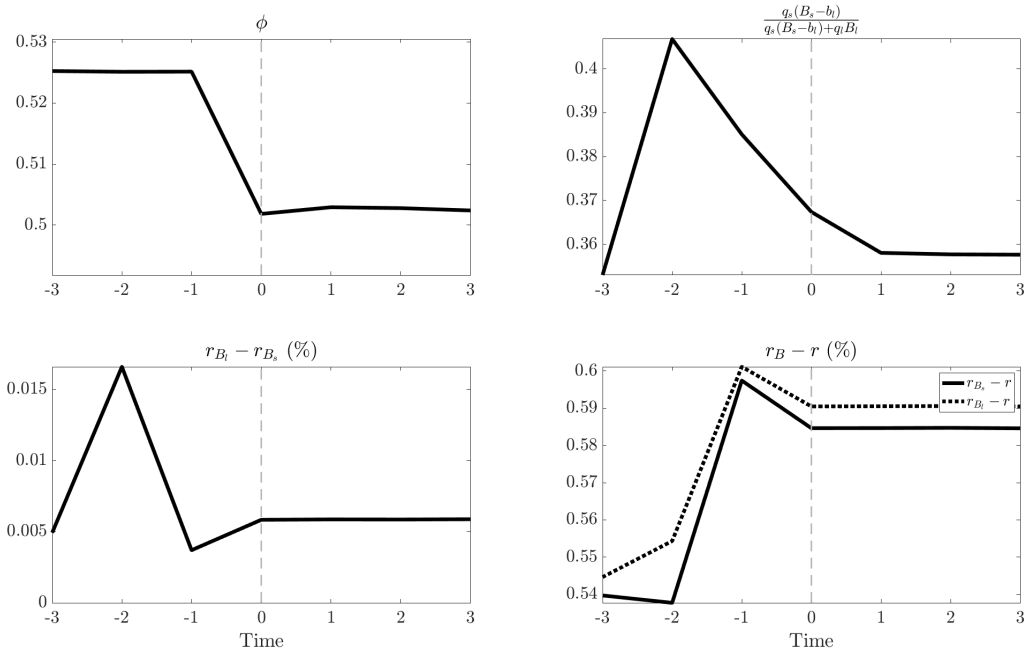


Note: The figure depicts the simulation of a one-period shock to the rate parameter of the distribution F. In period 0, the parameter is given by $\tilde{\eta} = 2$, while in all other periods it is given by the calibrated $\eta = 2.5 > \tilde{\eta}$. The shock is fully anticipated. In that logic, from period 0 onward, the model corresponds to the calibrated infinite horizon model. In the periods preceding period 0, the model is solved by backward induction.

Figure 3: One-period Shock, $\rho = 0$

We model the default-cost shock as a persistent regime switch. Initially, the economy is in the benchmark regime characterized by the calibrated rate parameter η of the default-cost distribution. We then assume an anticipated transition to a low-default-cost regime, $\tilde{\eta} < \eta$, of uncertain duration. The expected cost of default is lower in this regime, thus default becomes more likely. Thereafter, the low-default-cost regime persists with probability $\rho \in (1/2, 1]$, while with the counter probability the economy reverts permanently to the benchmark regime with parameter η. We consider first a permanent regime change ($\rho = 1$) and then a persistent one ($\rho = 0.9$). Comparing the two cases isolates the role of the expected duration of high default risk in the choice between short- and long-term debt. In both case we assume $\tilde{\eta} = 2$.

Figure 4 illustrates the response to a permanent regime switch ($\rho = 1$). Each panel



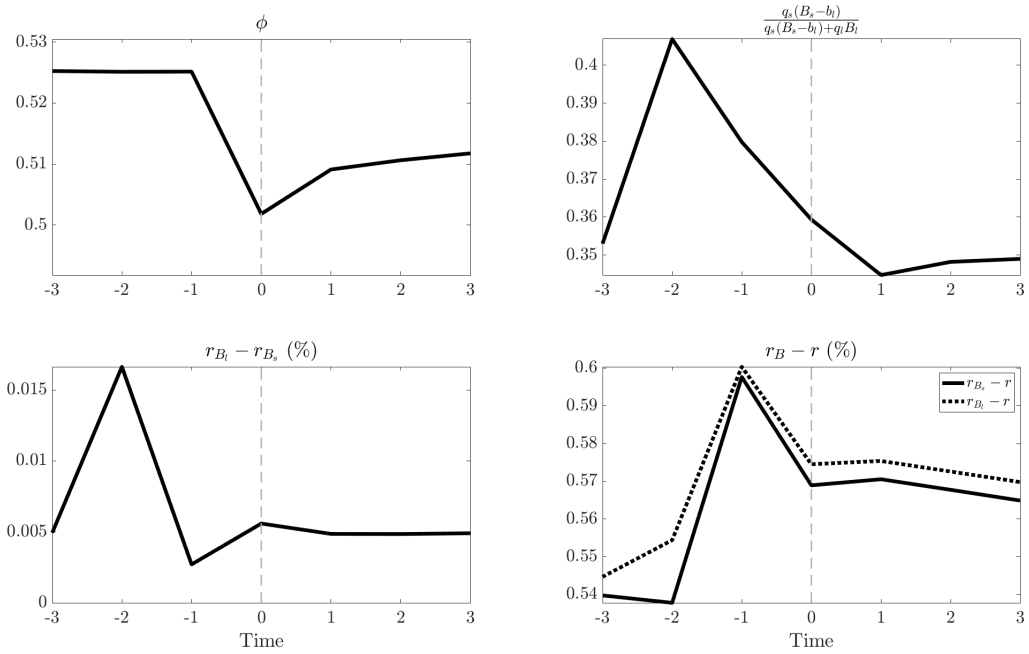
Note: The figure depicts the simulation of a permanent shock to the rate parameter of the distribution F. In period 0 and all periods onward, the parameter is given by $\tilde{\eta} = 2 < 2.5 = \eta$. The shock is fully anticipated. In the periods preceding period 0, the model is solved by backward induction.

Figure 4: Permanent Shock, $\rho = 1$

plots the expected value of the corresponding variable. In period 0, the economy switches permanently to a regime with a lower expected default cost, characterized by $\tilde{\eta} < \eta$ (top-left panel). In period -2, this regime switch becomes common knowledge and because long-term debt issued in period -2 matures in period 0, its holders are exposed to the higher default risk induced by the anticipated regime change. As a result, the long-term spread rises already in period -2, while the short-term spread remains largely unchanged, causing the term spread to increase (bottom panels). In response, the government shifts debt issuance toward short-term debt (top-right panel).

In period -1, issued short-term debt will mature in period 0 and long-term debt in period 1, where default risk is elevated in both periods. This causes short- and long-term spreads to increase by similar amounts. As a result, the term spread remains positive and the shift toward short-term issuance moderates. From period 0 onward, the economy converges to the new high-default-risk stationary equilibrium, characterized by persistently higher spreads, a larger term spread, and a higher revenue share of short term debt issuance.

Figure 4 depicts the response to a permanent shock ($\rho = 1$). Up to period -2, the dynamics are identical to those under a one-period shock. Since long-term debt issued in period -2 matures in period 0, the term spread rises and issuance shifts toward shorter



Note: The figure depicts the simulation of a persistent shock to the rate parameter of the distribution F. In period 0, the parameter is given by $\tilde{\eta} = 2$, while in the previous periods it is given by the calibrated $\eta = 2.5 > \tilde{\eta}$ and in the following periods the transition from $\tilde{\eta}$ to η is given by the probability $\rho = 0.9$. The transition back to η is absorbing. The shock is fully anticipated. In that logic, from period 0 onwards,the model corresponds to the calibrated infinite horizon model augmented with the aforementioned transition probabilities. In the periods preceding period 0, the model is solved by backward induction.

Figure 5: Persistent Shock, $\rho = 0.9$

maturities. The dynamics diverge in period -1: long-term debt issued then matures in period 1, when default risk remains elevated, causing long- and short-term spreads to increase by similar amounts. As a result, the yield curve remains upward sloping and the shift toward short-term issuance is modest. From period 0 onward, the economy converges to the new high-risk steady state, characterized by persistently higher spreads, a larger term spread, and a higher share of short-term debt issuance.

Finally, Figure 5 depicts the response to a persistent shock ($\rho = 0.9$). Because the economy exits the low-default-cost regime only gradually, the dynamics before period 0 resemble those under a permanent shock. In period -2, both the term spread and the share of short-term issuance increase. In period -1, long-term debt maturing in period 1 remains exposed to elevated default risk with positive probability, causing both spreads to rise without an inversion of the yield curve. Consequently, the decline in the share of short-term issuance is similar to that in the permanent-shock case. After period 0, spreads settle between the responses observed under the one-period and permanent shocks. The term spread returns close to its pre-shock level, while the maturity structure gradually converges back to its initial steady state. Thus, the persistent shock combines the anticipation effects

of the permanent shock with the gradual normalization observed under the one-period shock.

5 Empirical Analysis

In this section, we test the empirical predictions of our theory. The model predicts that a decline in the expected cost of default raises sovereign bond spreads, with stronger effects at longer maturities, and induces governments to shift debt issuance toward shorter maturities. In the empirical analysis, we proxy for declines in the expected cost of default using electoral support for anti-austerity parties measured in opinion polls. We classify debt as short- or long-term according to whether it matures before or after the next national election.

5.1 Data

Our empirical analysis combines end-of-the-week sovereign yield data, weekly sovereign issuance volumes measured at market value, political polling data, and information on national elections. The sample covers January 2013 to December 2024, with country-specific starting dates reflecting differences in polling data availability. The sample covers Greece, Ireland, Italy, Portugal, and Spain, the euro-area economies most severely affected by the European sovereign debt crisis. The dataset is organized at a weekly frequency, with sovereign yields and issuance volumes distinguished by country and maturity, and polling observations by country and political party. The harmonization of data frequencies and sample coverage across sources is outlined in Appendix E.1.

Political polling data were provided to us by *POLITICO Poll of Polls*. We focus on parties that opposed fiscal austerity during and after the European sovereign debt crisis, some of which also expressed reservations about sovereign debt repayment. The selected parties are listed in Table 2. For countries with multiple relevant parties, we sum their polling support into a country-level measure of anti-austerity support. Appendix E.2 describes the underlying polling data and estimation of party support, alongside the criteria for party selection and the aggregation procedure.

Table 2: Anti-austerity parties included in the analysis

Country	Parties
Greece	SYRIZA, Anexartiti Ellines (ANEL), Golden Dawn (XA)
Ireland	Solidarity–People Before Profit (SPBP)
Italy	Lega Nord, Movimento Cinque Stelle (M5S)
Portugal	Bloco de Esquerda (BE)
Spain	Podemos

5.2 Yield Spreads: Short- vs. Long-Maturity Bonds

Our analysis begins by examining the relationship between electoral support for anti-austerity parties and sovereign yield spreads across bond maturities, using the zero-coupon yield spreads described in Appendix E.3. For country c, week t, and bond maturity j, we estimate

$$s^j_{c,t} = \beta_1 \text{Long}^j_{c,t} + \beta_2 \text{Poll}_{c,t} + \beta_3 \left(\text{Poll}_{c,t} \times \text{Long}^j_{c,t}\right) + \alpha_c + \lambda_t + \varepsilon^j_{c,t},$$

where $s^j_{c,t}$ denotes the sovereign yield spread relative to Germany at the corresponding maturity and $\text{Poll}_{c,t}$ measures polling support for anti-austerity parties net of the mean support in the sample. To distinguish between short- and long-maturity bonds, we introduce a maturity threshold h, above which bonds are classified as long-maturity. The indicator $\text{Long}_{j,h}$ equals one if maturity j is at least h years and zero otherwise. We estimate the specification separately for thresholds $h \in \{1, 2, 3, 4\}$ years.[9] All specifications include country fixed effects α_c and week fixed effects λ_t, with Driscoll–Kraay standard errors.[10] We first estimate the specification for the full panel, comprising Greece, Ireland, Italy, Portugal, and Spain.

The coefficient β_1 captures the difference in yield spreads between long- and short-maturity bonds when polling support is at its sample mean. Changes in polling support are associated with short-maturity spreads through β_2 and with long-maturity spreads through $\beta_2 + \beta_3$. Our main coefficient of interest is therefore β_3, which captures the differential association across maturities.

Table 3 reports the results for the full panel. The long-maturity coefficient is positive and statistically significant across all thresholds, indicating that long-maturity bonds exhibit yield spreads approximately 81 to 101 bp higher than short-maturity bonds at average polling support. Higher support for anti-austerity parties is also associated with higher sovereign yield spreads. A ten-percentage-point increase in polling support is associated with an increase of approximately 89 to 107 bp in short-maturity spreads, depending on the maturity threshold. The interaction coefficient is positive and statistically significant for the one- and two-year thresholds, indicating an additional increase in long-maturity spreads of approximately 10 and 18 bp, respectively, for a ten-percentage-point increase in polling support. For the three- and four-year thresholds, the interaction becomes small and statistically insignificant. Thus, the full panel provides only limited evidence of a systematically stronger association at longer maturities.

[9] We consider different thresholds to assess the sensitivity of the results to the maturity classification. The maximum threshold of four years corresponds to the regular parliamentary term in Greece, Portugal, and Spain, while Italy and Ireland allow terms of up to five years.

[10] Details on the lag-length selection for Driscoll–Kraay standard errors and their implementation across statistical software packages are provided in Appendix H.

Table 3: Panel estimates

	(1) Yield Spread (≥ 1Y)	(2) Yield Spread (≥ 2Y)	(3) Yield Spread (≥ 3Y)	(4) Yield Spread (≥ 4Y)
Long Mat.	1.012***	0.969***	0.881***	0.807***
	[0.043]	[0.034]	[0.030]	[0.029]
Poll	0.089***	0.097***	0.104***	0.107***
	[0.012]	[0.013]	[0.014]	[0.014]
Long Mat. × Poll	0.018***	0.010***	0.002	-0.003
	[0.004]	[0.004]	[0.004]	[0.004]
Obs.	40695	40695	40695	40695
R^2	0.673	0.678	0.678	0.676

Notes: The table reports OLS estimates of raw sovereign yield spreads. Poll is the sum of support for the relevant anti-austerity parties within each country and is demeaned by subtracting its sample mean. All specifications include Country and Time fixed effects. Driscoll–Kraay standard errors with 5 lags are reported in brackets. Significance levels: * 10%, ** 5%, *** 1%.

However, the country-specific estimates reported in Appendix F reveal substantial heterogeneity, particularly for Greece, where the interaction coefficient turns negative and statistically significant at longer maturity thresholds. This contrasting pattern, together with the large movements in Greek sovereign yield spreads during the European sovereign debt crisis, motivates reestimating the pooled specification excluding Greece.

Table 4: Panel estimates without Greece

	(1) Yield Spread (≥ 1Y)	(2) Yield Spread (≥ 2Y)	(3) Yield Spread (≥ 3Y)	(4) Yield Spread (≥ 4Y)
Long Mat.	0.865***	0.907***	0.898***	0.872***
	[0.035]	[0.035]	[0.030]	[0.026]
Poll	0.013**	0.014**	0.015***	0.016***
	[0.006]	[0.006]	[0.005]	[0.005]
Long Mat. × Poll	0.011***	0.010***	0.010***	0.009***
	[0.001]	[0.001]	[0.001]	[0.001]
Obs.	31785	31785	31785	31785
R^2	0.576	0.632	0.665	0.678

Notes: The table reports OLS estimates of raw sovereign yield spreads. Poll is the sum of support for the relevant anti-austerity parties within each country and is demeaned by subtracting its sample mean. All specifications include Country and Time fixed effects. Driscoll–Kraay standard errors with 5 lags are reported in brackets. Significance levels: * 10%, ** 5%, *** 1%.

As shown in Table 4, excluding Greece substantially reduces the estimated short-maturity effect to approximately 13 to 16 bp for a ten-percentage-point increase in polling support. At the same time, the interaction coefficient remains positive and significant across all ma-

turity thresholds. A ten-percentage-point increase in polling support is associated with an additional 9 to 11 bp in long-maturity spreads relative to short-maturity spreads. These findings suggest that Greece drives much of the large average polling effect in the full panel, while its distinct maturity pattern weakens the differential association observed among the remaining countries, particularly at longer maturities.

5.3 Yield Spreads: Post-Election Maturities

The benchmark specification distinguishes between short- and long-term bonds using fixed maturity thresholds. However, this classification does not directly account for the electoral process through which political support may affect sovereign risk. In particular, whether a bond matures before or after the next election determines its exposure to the policies of a potentially new government. We therefore introduce a time-varying post-election indicator that identifies maturities extending beyond the next national election, allowing us to examine whether the relationship between anti-austerity party support and sovereign yield spreads differs with electoral exposure. We estimate

$$s^j_{c,t} = \beta_1 \text{PostElection}^j_{c,t} + \beta_2 \text{Poll}_{c,t} + \beta_3 \left(\text{Poll}_{c,t} \times \text{PostElection}^j_{c,t}\right) + \alpha_c + \lambda_t + \varepsilon^j_{c,t},$$

where $\text{PostElection}^j_{c,t}$ equals one if maturity j extends beyond the relevant election date. The level effect of bonds maturing after an election is indicated by β_1. The coefficient β_2 captures the association between polling support and spreads for pre-election maturities, while β_3 measures the additional association for post-election maturities. We consider two election-timing assumptions: one treating all realized election dates as fully anticipated, including early elections, and another updating the expected election date only after an early election is announced. The definitions of the two election-timing indicators and the differences between them are explained in Appendix E.4. As in the benchmark specification, we report pooled estimates with country and week fixed effects and country-specific estimates without fixed effects, using Driscoll–Kraay standard errors throughout.

Table 5 reports the results when all elections are treated as fully anticipated. The post-election indicator is positive and statistically significant across all specifications, indicating higher spreads for maturities extending beyond the next election. Consistent with the benchmark results, higher polling support for anti-austerity parties is generally associated with higher spreads. The interaction coefficient is positive and statistically significant in both pooled specifications, suggesting a stronger association between polling support and spreads for post-election maturities. In the full panel, a 10-percentage-point increase in polling support is associated with an increase in spreads of approximately 95 bp for pre-election

Table 5: Post-election maturity specification with fully anticipated elections

	(1) Pooled	(2) Pooled (No GR)	(3) Greece	(4) Spain	(5) Portugal	(6) Italy	(7) Ireland
Post-Elec.	0.831***	0.800***	0.587***	0.627***	1.019***	0.908***	0.327***
	[0.042]	[0.034]	[0.184]	[0.015]	[0.069]	[0.028]	[0.015]
Poll	0.095***	0.017***	0.275***	0.016***	0.084***	0.005	0.026*
	[0.010]	[0.006]	[0.024]	[0.003]	[0.021]	[0.004]	[0.015]
Post-Elec. × Poll	0.011**	0.008***	0.006	0.006**	0.028	0.001	0.022*
	[0.005]	[0.001]	[0.024]	[0.003]	[0.035]	[0.003]	[0.012]
Obs.	40695	31785	8910	7395	8910	8940	6540
R^2	0.671	0.602	0.593	0.314	0.117	0.295	0.302
Country & Time FE	Yes	Yes	No	No	No	No	No

Notes: The table reports OLS estimates of raw sovereign yield spreads. Poll is the sum of support for the relevant anti-austerity parties within each country and is demeaned by subtracting its sample mean. Post-Elec. equals one when maturity is on or after the next scheduled election. Driscoll–Kraay standard errors with 5 lags are reported in brackets. Significance levels: * 10%, ** 5%, *** 1%.

maturities. This is increased by 11 bp to 106 bp for post-election maturities. Excluding Greece, these effects are considerably smaller, amounting to 17 and 25 bp, respectively. Thus, while Greece contributes substantially to the overall magnitude, the stronger response of post-election maturities remains statistically significant when Greece is excluded.

Table 6: Post-election maturity specification with early-election surprises

	(1) Pooled	(2) Pooled (No GR)	(3) Greece	(4) Spain	(5) Portugal	(6) Italy	(7) Ireland
Post-Elec.	0.698***	0.823***	0.222	0.644***	1.160***	0.915***	0.332***
	[0.051]	[0.031]	[0.148]	[0.015]	[0.073]	[0.027]	[0.012]
Poll	0.110***	0.018***	0.336***	0.019***	0.091***	0.007	0.029**
	[0.014]	[0.006]	[0.028]	[0.003]	[0.019]	[0.005]	[0.014]
Post-Elec. × Poll	-0.006	0.007***	-0.075***	0.005*	0.013	-0.002	0.018*
	[0.005]	[0.001]	[0.016]	[0.003]	[0.034]	[0.003]	[0.010]
Obs.	40695	31785	8910	7395	8910	8940	6540
R^2	0.665	0.624	0.597	0.389	0.167	0.308	0.342
Country & Time FE	Yes	Yes	No	No	No	No	No

Notes: The table reports OLS estimates of raw sovereign yield spreads. Poll is the sum of support for the relevant anti-austerity parties within each country and is demeaned by subtracting its sample mean. Post-Elec. equals one when maturity is on or after the early-adjusted election date. Driscoll–Kraay standard errors with 5 lags are reported in brackets. Significance levels: * 10%, ** 5%, *** 1%.

Table 6 presents results for early-election surprises. The post-election indicator remains positive and statistically significant in both pooled specifications. However, the interaction

coefficient becomes negative and statistically insignificant in the full panel, while remaining positive and statistically significant when Greece is excluded. In this election specification, a 10-percentage-point increase in polling support is associated with approximately 18 bp higher spreads for pre-election maturities and 25 bp for post-election maturities, closely resembling the fully anticipated results. The difference in the full panel is largely driven by Greece, where the interaction is negative and significant. For Greece, the same increase in polling support is associated with 336 bp higher spreads for pre-election maturities, compared to 261 bp for post-election maturities. This is consistent with the previous benchmark results, which also highlighted Greece as a distinct case in the relationship between political support and sovereign yield spreads.

Overall, the results extend the benchmark findings by showing that the relationship between political support and sovereign yield spreads also varies with exposure to the electoral process. While the fully anticipated specification suggests stronger responses for post-election maturities, accounting for unexpected elections leaves the positive relationship largely unchanged outside Greece, where the interaction becomes negative and statistically significant.

5.4 Bond Issuance: Short- vs. Long-Maturity Bonds

The previous analysis examined how electoral support for anti-austerity parties is associated with sovereign yield spreads, capturing how political risk is priced in sovereign debt markets. We now turn to sovereign debt issuance to investigate whether this relationship also extends to governments' borrowing decisions. In particular, we examine whether electoral support is associated with changes in the volume of newly issued debt and its maturity composition. These adjustments are important as they determine governments' exposure to refinancing risk and the extent to which current borrowing conditions are locked in over time.

We aggregate weekly bond issuance into short- and long-term maturity categories using thresholds of one, two, three, and four years. Issuance is normalized by its respective sample average to account for differences across countries and maturity categories. The construction of market-value issuance and its normalization across countries and maturity groups are described in Appendix E.5.

We estimate the relationship between electoral support and normalized issuance using Poisson Pseudo-Maximum Likelihood (PPML), which accommodates non-negative issuance volumes, including observations without issuance. The specification is given by

$$\log \mathbb{E}\left[\widetilde{I}^{j}_{c,b,t} \mid X_{c,b,t}\right] = \beta_1 \text{Long}^{j}_{b} + \beta_2 \text{Poll}_{c,t} + \beta_3 \left(\text{Poll}_{c,t} \times \text{Long}^{j}_{b}\right) + \alpha_c + \lambda_t,$$

where $\widetilde{I}^{j}_{c,b,t}$ denotes normalized weekly issuance for country c, maturity category b, and week t, with j indicating the respective maturity threshold. $Long^{j}_{b}$ identifies long-term debt, while $Poll_{c,t}$ represents demeaned electoral support for anti-austerity parties measured at the beginning of the issuance week. The interaction allows the relationship between electoral support and issuance to differ across maturity categories. All specifications include country and week fixed effects, with Driscoll–Kraay standard errors (see Appendix H for details on lag determination).

For economic interpretation, we report average marginal effects (AMEs) of a one-percentage-point increase in electoral support on expected normalized issuance.[11] Table 7 presents the AMEs for the pooled sample. The underlying PPML estimates are reported in Appendix Table G.9.

Table 7: Average Marginal Effects - Pooled - Maturity Cutoffs

	(1) $\geq$ 1 Year	(2) $\geq$ 2 Years	(3) $\geq$ 3 Years	(4) $\geq$ 4 Years
Short maturity	-0.008	-0.014**	-0.014**	-0.016***
	[0.006]	[0.006]	[0.006]	[0.006]
Long maturity	-0.015**	-0.021***	-0.021***	-0.023***
	[0.007]	[0.007]	[0.007]	[0.007]
Difference: long - short	-0.008	-0.007	-0.007	-0.007
	[0.005]	[0.005]	[0.005]	[0.005]
Obs.	2659	2659	2659	2659
R^2	0.130	0.143	0.143	0.147

Notes: The table reports average marginal effects of Poll on expected normalized issuance. Poll is the sum of support for the relevant anti-austerity parties within each country and is demeaned by subtracting its sample mean. The difference is the long/post-election AME minus the short/pre-election AME. For PPML, R^2 denotes the pseudo-R^2. All specifications include Country and Time fixed effects. Driscoll–Kraay standard errors with lag 5 are reported in brackets. Significance levels: * 10%, ** 5%, *** 1%.

Across all maturity thresholds, higher electoral support is associated with lower short- and long-term issuance. The relationship is statistically significant for long-term issuance at every threshold. For short-term issuance, it is significant at thresholds of two years and above. A ten-percentage-point increase in support is associated with a reduction of $8-16\%$ of average weekly short-term issuance, compared to $1.5-2.3\%$ for long-term issuance. The long-minus-short difference is consistently negative but not statistically significant. Its estimated magnitude nevertheless suggests an economically meaningful difference. For a ten-

[11] PPML coefficients describe proportional changes in expected issuance, whereas AMEs express changes in expected normalized issuance. Due to the normalization, an AME of -0.02 corresponds to a reduction equivalent to 2% of average weekly issuance in the respective maturity category.

percentage-point increase in support, the AMEs imply a roughly $7-8$ percentage-point larger decline in long-term issuance, measured relative to each category's mean. The estimates therefore suggest a potentially substantial shift away from long-term borrowing, although the differential response remains statistically insignificant.

Country-specific estimates are presented in Appendix G. Next, we distinguish between debt maturing before and after the next election to examine whether the relationship between electoral support and issuance depends on election timing.

5.5 Bond Issuance: Post-Election Maturities

Having examined differences in issuance across short- and long-term maturities, we next investigate whether issuance responses to electoral support depend on whether bonds mature before or after the next election. As in the yield-spread analysis in section 5.3, we distinguish between fully anticipated elections and early-election surprises, classifying bonds according to whether they mature before, on, or after the relevant election date.

We retain the PPML framework and normalized issuance measure introduced in the previous subsection, replacing the long-maturity indicator with the election-based maturity classification. The specification is given by

$$\log \mathbb{E}\left[\widetilde{I}^{j}_{c,b,t} \mid X_{c,b,t}\right] = \beta_1 \text{PostElection}^{j}_{c,t} + \beta_2 \text{Poll}_{c,t} + \beta_3 \left(\text{Poll}_{c,t} \times \text{PostElection}^{j}_{c,t}\right) + \alpha_c + \lambda_t,$$

where $\text{PostElection}^{j}_{c,t}$ equals one if the bond matures on or after the relevant election date and zero otherwise. We estimate the specification separately for the two election-timing approaches, either treating all elections as fully anticipated or accounting for early-election surprises once an early election is announced or triggered. As before, the pooled specifications include country and week fixed effects, while country-specific specifications are estimated without them. Driscoll–Kraay standard errors are used throughout.

We again report average marginal effects (AMEs) of a one-percentage-point increase in electoral support, separately for pre- and post-election maturities, together with their difference. The underlying PPML estimates are reported in Appendix Tables G.20 and G.21.

Table 8 presents the results when all elections are treated as fully anticipated. The pooled AMEs imply declines of approximately 4% and 5% of average weekly pre- and post-election issuance for a ten-percentage-point increase in support. Neither effect nor their difference is statistically significant. Portugal exhibits a substantial tilt toward debt maturing before the election. For the same increase in support, the response of pre-election issuance exceeds that of post-election issuance by approximately 175 percentage points, with each response mea-

Table 8: Average Marginal Effects of Bond Issuance with Fully Anticipated Elections

	(1) Pooled	(2) Greece	(3) Spain	(4) Portugal	(5) Italy	(6) Ireland
Pre-elec. mat.	-0.004	0.012*	-0.021	0.151***	-0.003	-0.039
	[0.005]	[0.007]	[0.016]	[0.057]	[0.004]	[0.139]
Post-elec. mat.	-0.005	0.006	-0.007	-0.024	-0.005	-0.183
	[0.007]	[0.015]	[0.009]	[0.045]	[0.004]	[0.116]
Diff.: post - pre	-0.001	-0.006	0.014	-0.175**	-0.002	-0.144
	[0.005]	[0.019]	[0.021]	[0.074]	[0.007]	[0.172]
Obs.	2659	542	490	594	596	437
R^2	0.128	0.002	0.002	0.009	0.001	0.002
Country & Time FE	Yes	No	No	No	No	No

Notes: The table reports average marginal effects of Poll on expected normalized issuance. Poll is the sum of support for the relevant anti-austerity parties within each country and is demeaned by subtracting its sample mean. The difference is the post-election AME minus the pre-election AME. For PPML, R^2 denotes the pseudo-R^2. Driscoll–Kraay standard errors with lag 5 are reported in brackets. Significance levels: * 10%, ** 5%, *** 1%.

sured relative to its mean issuance. This difference is statistically significant and primarily reflects increased pre-election borrowing. Differences remain statistically insignificant in the other countries.

Table 9: Average Marginal Effects of Bond Issuance with Early-Election Surprises

	(1) Pooled	(2) Greece	(3) Spain	(4) Portugal	(5) Italy	(6) Ireland
Pre-elec. mat.	-0.009*	0.025***	-0.002	0.129**	-0.003	-0.122
	[0.005]	[0.005]	[0.012]	[0.052]	[0.004]	[0.134]
Post-elec. mat.	-0.015**	-0.036**	-0.015	-0.020	-0.005	-0.164
	[0.006]	[0.015]	[0.010]	[0.047]	[0.004]	[0.117]
Diff.: post - pre	-0.006	-0.061***	-0.013	-0.149**	-0.002	-0.042
	[0.005]	[0.016]	[0.018]	[0.072]	[0.007]	[0.169]
Obs.	2659	542	490	594	596	437
R^2	0.132	0.018	0.001	0.006	0.001	0.003
Country & Time FE	Yes	No	No	No	No	No

Notes: The table reports average marginal effects of Poll on expected normalized issuance. Poll is the sum of support for the relevant anti-austerity parties within each country and is demeaned by subtracting its sample mean. The difference is the post-election AME minus the pre-election AME. For PPML, R^2 denotes the pseudo-R^2. Driscoll–Kraay standard errors with lag 5 are reported in brackets. Significance levels: * 10%, ** 5%, *** 1%.

Table 9 reports the results accounting for early-election surprises. The pooled AMEs are

negative and statistically significant for both maturity categories, although their difference remains insignificant. Greece and Portugal exhibit a significant tilt toward debt maturing before the election. For a ten-percentage-point increase in support, pre-election issuance responds approximately 61 and 149 percentage points more positively than post-election issuance, respectively, with each response measured relative to its mean issuance. The estimates suggest a substantial shift toward borrowing due before the adjusted election date. For the remaining countries, the differences point in the same direction but are statistically insignificant.

Overall, the pooled estimates indicate a negative relationship between electoral support and issuance in both maturity categories. The decline is generally stronger for debt maturing after the next election, although the pooled differences are imprecisely estimated. The country-specific results, particularly for Greece and Portugal, provide stronger evidence of this differential response. Taken together, the findings suggest maturity-dependent adjustments in sovereign borrowing, with a tendency toward debt maturing before the next election.

5.6 Bond Issuance: Average Maturity

While the previous analysis examines how electoral support for anti-austerity parties affects issuance volumes across maturity categories, we now turn to the maturity composition of newly issued debt. The PPML estimates suggest that longer-maturity issuance tends to respond more negatively to increases in anti-austerity support, although differences across maturity categories are generally not statistically significant. To investigate maturity composition more directly, we construct an issuance-weighted average maturity over a rolling window of 26 weeks, as described in Appendix E.6. This measure captures changes in the overall maturity structure of issuance rather than differences between discrete maturity categories. We estimate the relationship between polling support and average maturity using the following OLS specification

$$\overline{M}_{c,t}^{(26)} = \beta_1 \text{Poll}_{c,t} + \alpha_c + \lambda_t + \varepsilon_{c,t},$$

where $\overline{M}_{c,t}^{(26)}$ denotes the issuance-weighted average maturity over the trailing 26 weeks. The specification includes country and week fixed effects, with Driscoll–Kraay standard errors.

Table 10 reports the OLS estimates of the relationship between polling support for anti-austerity parties and the trailing six-month average maturity. In the full pooled sample, the estimated coefficient is negative but statistically insignificant. Excluding Italy, however, the coefficient becomes positive and statistically significant, suggesting an increase in average

Table 10: Trailing Six-Month Average Maturity

	(1) Pooled	(2) Pooled (No IT)	(3) Greece	(4) Spain	(5) Portugal	(6) Italy	(7) Ireland
Poll	-0.114	-0.736**	-1.742***	-0.670**	-3.987***	0.322***	-10.857**
	[0.184]	[0.292]	[0.169]	[0.262]	[1.043]	[0.097]	[4.454]
Obs.	2647	2051	542	490	582	596	437
R^2	0.773	0.802	0.460	0.073	0.088	0.109	0.048
Ctry & Time FE	Yes	Yes	No	No	No	No	No

Notes: The table reports OLS estimates of trailing six-month average maturity. Poll is the sum of support for the relevant anti-austerity parties within each country and is demeaned by subtracting its sample mean. Average maturity is constructed from raw issuance using exact maturities over the current and previous 25 weeks before the poll sample is imposed. Driscoll–Kraay standard errors with lag 5 are reported in brackets. Significance levels: * 10%, ** 5%, *** 1%.

maturity of approximately 0.74 months following a one-percentage-point increase in polling support.

The country-specific estimates reveal substantial heterogeneity. For Greece, Spain, Portugal, and Ireland, higher polling support is associated with a statistically significant reduction in average maturity. A one-percentage-point increase in polling support corresponds to a reduction of approximately 1.74 months in Greece, 0.67 months in Spain, 3.99 months in Portugal, and 10.86 months in Ireland. Italy exhibits the opposite relationship, with a statistically significant increase of approximately 0.32 months. Overall, the country-specific results suggest that stronger support for anti-austerity parties is associated with a shift toward shorter debt maturities in four of the five countries, although this pattern is not reflected consistently in the pooled estimates.

References

Aguiar, M. and M. Amador (2020): "Self-Fulfilling Debt Dilution: Maturity and Multiplicity in Debt Models," *American Economic Review*, 110, 2783–2818.

Aguiar, M., M. Amador, H. Hopenhayn, and I. Werning (2019): "Take the Short Route: Equilibrium Default and Debt Maturity," *Econometrica*, 87, 423–462.

Aguiar, M. and G. Gopinath (2006): "Defaultable Debt, Interest Rates and the Current Account," *Journal of International Economics*, 69, 64–83.

Alesina, A. and G. Tabellini (1990): "A Positive Theory of Fiscal Deficits and Government Debt," *Review of Economic Studies*, 57, 403–414.

Andreasen, E., G. Sandleris, and A. Van der Ghote (2019): "The political economy of sovereign defaults," *Journal of Monetary Economics*, 104, 23–36.

Angeletos, G.-M. (2002): "Fiscal Policy with Noncontingent Debt and the Optimal Maturity Structure," *Quarterly Journal of Economics*, 117, 1105–1131.

Arce, F., J. Morgan, and N. Werquin (2024): "Tax Revolts and Sovereign Defaults," *Federal Reserve Bank of Chicago Working Papers*.

Arellano, C. (2008): "Default Risk and Income Fluctuations in Emerging Economies," *American Economic Review*, 98, 690–712.

Arellano, C. and A. Ramanarayanan (2012): "Default and the Maturity Structure in Sovereign Bonds," *Journal of Political Economy*, 120, 187–232.

Azzimonti, M. (2011): "Barriers to Investment in Polarized Societies," *American Economic Review*, 101, 2182–2204.

Azzimonti, M. and N. Mitra (2023): "Political constraints and sovereign default," *Journal of International Money and Finance*, 137, 102895.

Bai, Y., S. T. Kim, and G. Mihalache (2017): "The Payment Schedule of Sovereign Debt," *Economics Letters*, 161, 19–23.

Battaglini, M. and S. Coate (2008): "A Dynamic Theory of Public Spending, Taxation, and Debt," *American Economic Review*, 98, 201–36.

Broner, F. A., G. Lorenzoni, and S. L. Schmukler (2013): "Why Do Emerging Economies Borrow Short Term?" *Journal of the European Economic Association*, 11, 67–100.

Buera, F. and J. P. Nicolini (2004): "Optimal Maturity of Government Debt Without State Contingent Bonds," *Journal of Monetary Economics*, 51, 531–554.

Bussière, M. and C. Mulder (2000): "Political Instability and Economic Vulnerability," *International Journal of Finance and Economics*, 5, 309–330.

Chatterjee, S., M. Dotsey, and G. Gordon (2026): "Debt Maturity in the 18th Century: Were Consols Optimal?" .

Chatterjee, S. and B. Eyigungor (2012): "Maturity, Indebtedness, and Default Risk," *American Economic Review*, 102, 2674–2699.

——— (2015): "A Seniority Arrangement for Sovereign Debt," *American Economic Review*, 105, 3740–3765.

Cotoc, J., A. Johri, and C. Sosa-Padilla (2025): "Sovereign Spreads and the Political Leaning of Nations," *International Economic Review*, 66, 687–709.

Cukierman, A. and A. H. Meltzer (1989): "A Political Theory of Government Debt and Deficits in a Neo-Ricardian Framework," *The American Economic Review*, 79, 713–732.

Debortoli, D., R. Nunes, and P. Yared (2017): "Optimal Time-Consistent Government Debt Maturity," *Quarterly Journal of Economics*, 132, 55–102.

Dovis, A., M. Golosov, and A. Shourideh (2016): "Political Economy of Sovereign Debt: A Theory of Cycles of Populism and Austerity," *NBER Working Paper*.

Eaton, J. and M. Gersovitz (1981): "Debt with Potential Repudiation: Theoretical and Empirical Analysis," *Review of Economic Studies*, 48, 289–309.

Faraglia, E., A. Marcet, and A. Scott (2010): "In Search of a Theory of Debt Management," *Journal of Monetary Economics*, 57, 821–836.

Farhi, E. and I. Werning (2007): "Inequality and Social Discounting," *Journal of Political Economy*, 115, 365–402.

Greenwood, J., Z. Hercowitz, and G. W. Huffman (1988): "Investment, Capacity Utilization, and the Real Business Cycle," *The American Economic Review*, 78, 402–417.

Hassler, J., P. Krusell, K. Storesletten, and F. Zilibotti (2005): "The dynamics of government," *Journal of Monetary Economics*, 52, 1331–1358.

Hassler, J., J. V. Rodríguez Mora, K. Storesletten, and F. Zilibotti (2003): "The Survival of the Welfare State," *American Economic Review*, 93, 87–112.

Hatchondo, J. C. and L. Martinez (2009): "Long-duration Bonds and Sovereign Defaults," *Journal of International Economics*, 79, 117–125.

Hatchondo, J. C., L. Martinez, and F. Roch (2020): "Constrained Efficient Borrowing with Sovereign Default Risk," *IMF Working Paper*.

Hatchondo, J. C., L. Martinez, and C. Sosa-Padilla (2016): "Debt Dilution and Sovereign Default Risk," *Journal of Political Economy*, 124, 1383–1422.

Hermann, T. and A. Scholl (2023): "The Political Economy of Domestic and External Sovereign Debt," *University of Konstanz*.

Herrera, H., G. Ordoñez, and C. Trebesch (2019): "Political Booms, Financial Crises," *Journal of Political Economy*, 128.

Kiiashko, S. (2022): "Optimal Time-Consistent Government Debt Maturity Fiscal Policy and Default," *Journal of the European Economic Association*, 20, 2062–2097.

Klein, P., P. Krusell, and J.-V. Ríos-Rull (2008): "Time-Consistent Public Policy," *Review of Economic Studies*, 75, 789–808.

Krusell, P., B. Kuruscu, and A. A. Smith (2002): "Equilibrium Welfare and Government Policy with Quasi-geometric Discounting," *Journal of Economic Theory*, 105, 42–72.

——— (2010): "Temptation and Taxation," *Econometrica*, 78, 2063–2084.

Krusell, P. and J.-V. Rios-Rull (1999): "On the Size of U.S. Government: Political Economy in the Neoclassical Growth Model," *American Economic Review*, 89, 1156–1181.

Krusell, P. and A. A. Smith (2003): "Consumption–Savings Decisions with Quasi–Geometric Discounting," *Econometrica*, 71, 365–375.

Lucas, R. E. and N. L. Stokey (1983): "Optimal Fiscal and Monetary Policy in an Economy Without Capital," *Journal of Monetary Economics*, 12, 55–93.

Mateos-Planas, X., S. McCrary, J.-V. Rios-Rull, and A. Wicht (2026): "The Generalized Euler Equation and the Bankruptcy-Sovereign Default Problem," *University of Pennsylvania*.

Mateos-Planas, X., S. McCrary, J.-V. Ríos-Rull, and A. Wicht (2025): "Commitment in the Canonical Sovereign Default Model," *Journal of International Economics*, 157.

Müller, A., K. Storesletten, and F. Zilibotti (2016): "The Political Color of Fiscal Responsibility," *Journal of the European Economic Association*, 14, 252–302.

——— (2019): "Sovereign Debt and Structural Reforms," *American Economic Review*, 109, 4220–4259.

Niepelt, D. (2014): "Debt Maturity Without Commitment," *Journal of Monetary Economics*, 68, 37–54.

Perez, D. J. (2017): "Sovereign Debt Maturity Structure under Asymmetric Information," *Journal of International Economics*, 108, 243–259.

Prein, T. M. and A. Scholl (2021): "The Impact of Bailouts on Political Turnover and Sovereign Default Risk," *Journal of Economic Dynamics and Control*, 124.

Scholl, A. (2017): "The Dynamics of Sovereign Default Risk and Political Turnover," *Journal of International Economics*, 108, 37–53.

Song, Z. (2011): "The dynamics of inequality and social security in general equilibrium," *Review of Economic Dynamics*, 14, 613–635.

——— (2012): "Persistent Ideology and the Determination of Public Policy over Time," *International Economic Review*, 53, 175–202.

Song, Z., K. Storesletten, and F. Zilibotti (2012): "Rotten Parents and Disciplined Children: A Politico-Economic Theory of Public Expenditure and Debt," *Econometrica*, 80, 2785–2803.

Stangebye, Z. R. (2020): "Beliefs and long-maturity sovereign debt," *Journal of International Economics*, 127, 103381.

Trabandt, M. and H. Uhlig (2011): "The Laffer curve revisited," *Journal of Monetary Economics*, 58, 305–327.

Yared, P. (2010): "A Dynamic Theory of War and Peace," *Journal of Economic Theory*, 145, 1921–1950.